

World Congress of Families (WFC)

- Family is one of the most fundamental social institutions, and the language influential organizations used to talk about it can affect public opinions, cultural norms, and even political policies.
- World Congress of Families is a global conference network that brings influential people from around the world to discuss family issues.
- Speeches in WCF provide a way to study how social and political themes are expressed over time.

Research Question

How do the World Congress of Families speeches changes over time in their use of language related to religiosity, science, and political rhetoric?

Data

- 652 full speech transcripts from World Congress of Families conferences.
- Speeches span 10 conference years between 1997–2019.

Conference Year	Speech
1997	"On June 8, 1989, at the final discussion of the 45th United Nations Human Rights..."
1997	"I begin with what..."
1997	"What does it mean..."

Table 1. Overview of project data.

Example Anchor Sentences

	Positive	Negative
Religion	Religious tradition preserves the wisdom of the ages.	Logic dictates the necessary conclusions for a rational life.
Science	Independent verification by external research groups confirms the reproducibility of these results.	I don't understand the statistics, but the conclusion matches my beliefs so it's valid.
Politics	The trade agreement projects a 3.2% GDP increase based on tariff reduction schedules and market access provisions.	They're using fear tactics to scare voters unlike our campaign, which presents the genuine serious threats.

Table 2. Positive and negative anchor examples for each topic.

Methods

- Built religion, science, and politics axes from positive vs. negative anchor sentence embeddings.
- Used a Sentence Transformer model to convert each sentence into a 384-dimensional vector.
- Computed each topic axis as the vector difference between averaged positive and negative anchors.
- Scored each speech sentence using cosine similarity to each topic axis.
- Used the 75th percentile of sentence scores per speech to capture strong topic signals while reducing neutral-sentence noise.
- Compared yearly speech-score distributions using violin plots and means to quantify the prevalence of topic justification within each conference year.
- Tested longitudinal trends with regression models and evaluated statistical significance using p-values.

Result

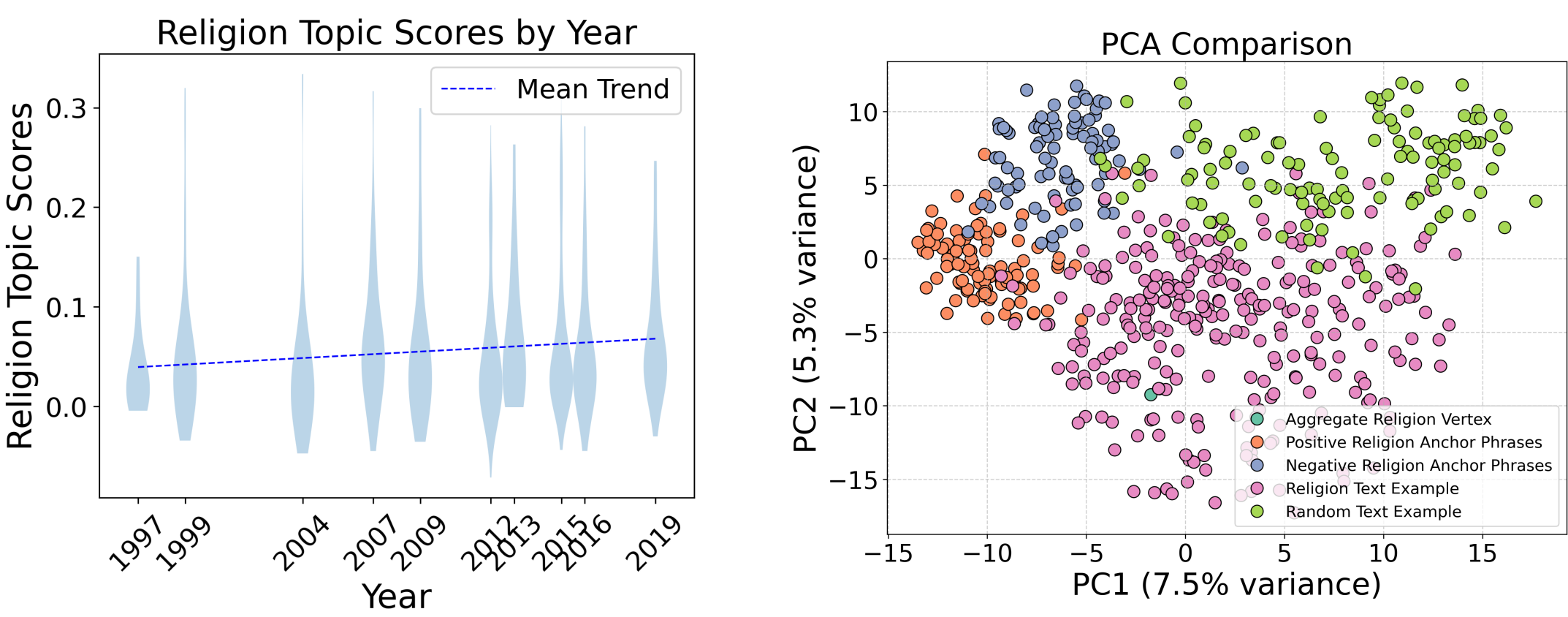


Figure 1. Religious trend and PCA. Slope = 0.0013, p-value = 0.0148, se = 0.0004.

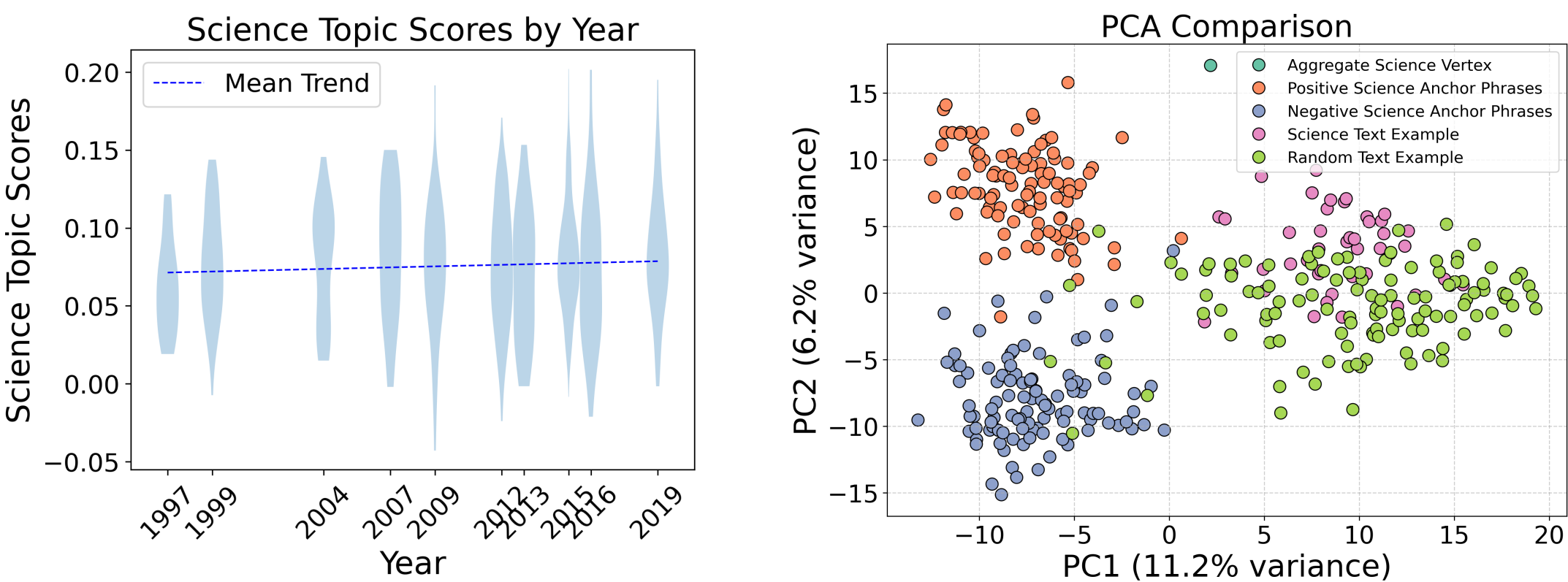


Figure 2. Science trend and PCA. Slope = -0.0010, p-value = 0.0139, se = 0.0003.

Result Cont.

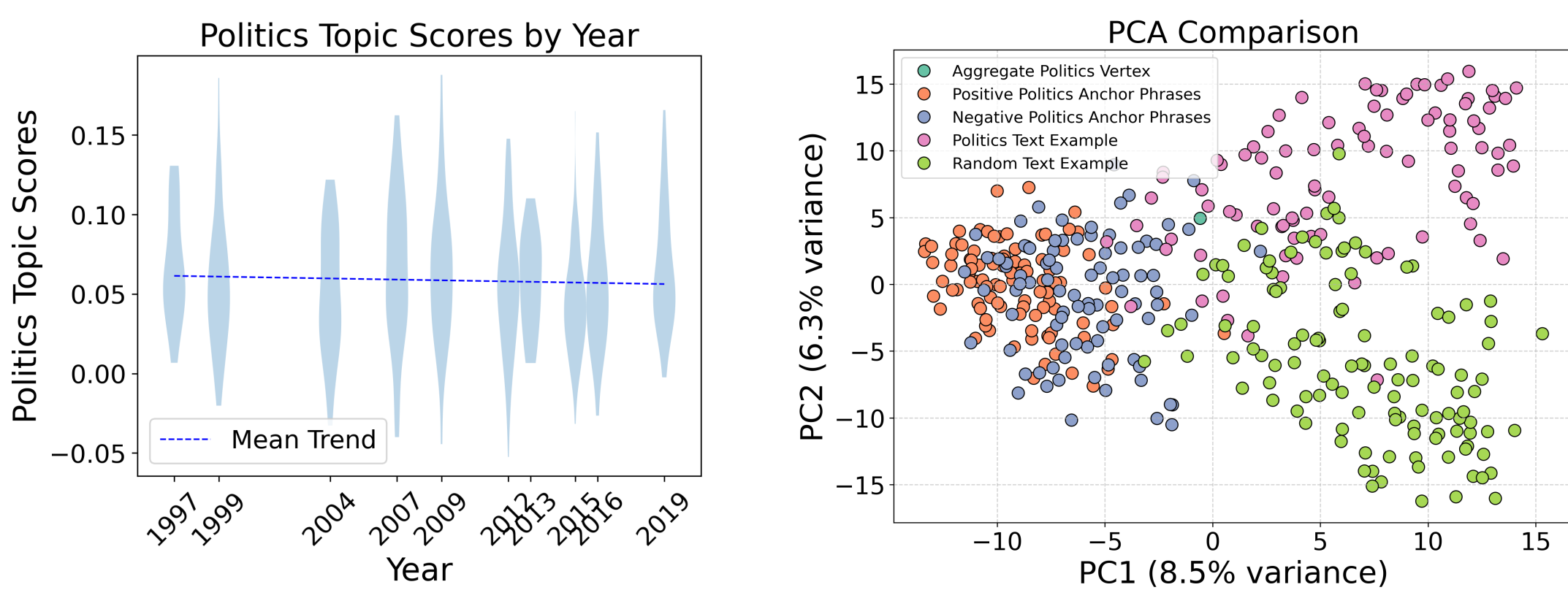


Figure 3. Politics trend and PCA. Slope = -0.0021, p-value = 0.0008, se = 0.0004.

Preliminary Conclusion

- Religion scores show a positive trend across years, suggesting increased use of religious framing in WCF speeches.
- Politics and science scores show negative trends, suggesting these topics were less consistently emphasized over time.
- PCS plots support that anchor axes capture meaningful separation between speech text.

Discussion

- Limited speech data in 1997 and 2013.
- Some yearly topic scores are unevenly distributed with long tails.
- Result sensitive to anchors.

References

[1] Rachel Kahn Best and Alina Arseniev-Koehler. The stigma of diseases: Unequal burden, uneven decline. *American Sociological Review*, 88(5):938–969, 2023.

[2] Kristopher Velasco and Jeffrey Swindle. Assembling god's "last best hope": The expanding reach of the world congress of families. *Mobilization: An International Quarterly*, 29(4):441–468, 2024.

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